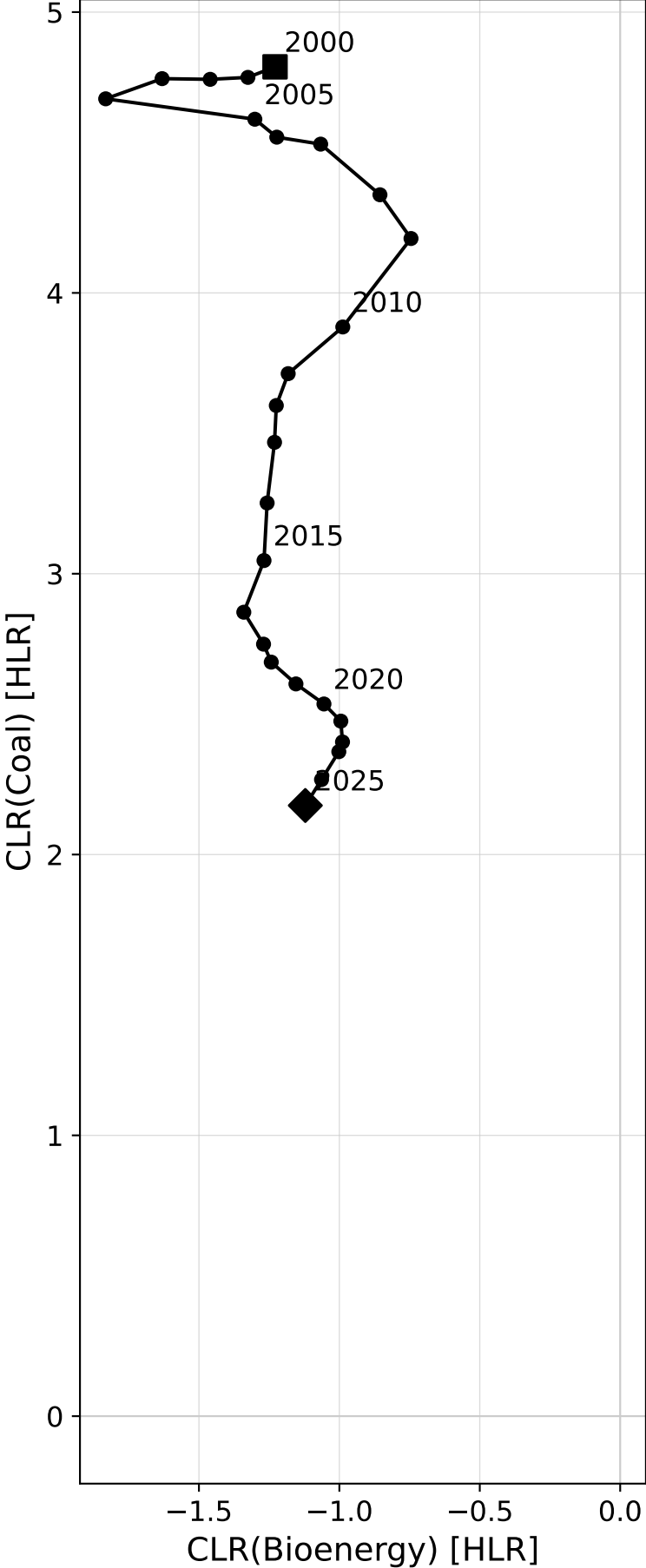


**SYSTEM COURSE PLOT — XY Plan View (CLR Space)**

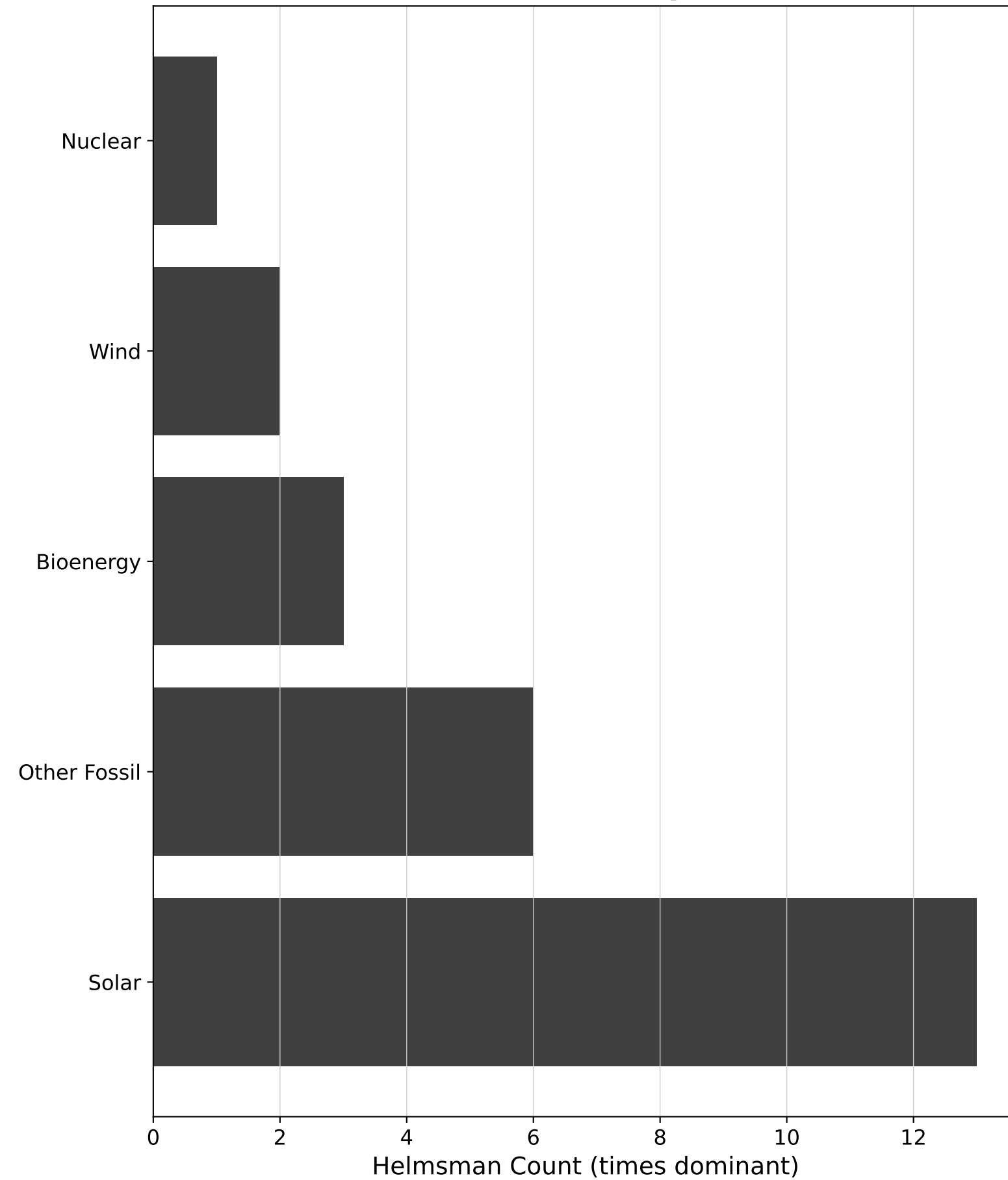


## STAGE 2 — Aitchison Distance Matrix (all year pairs)

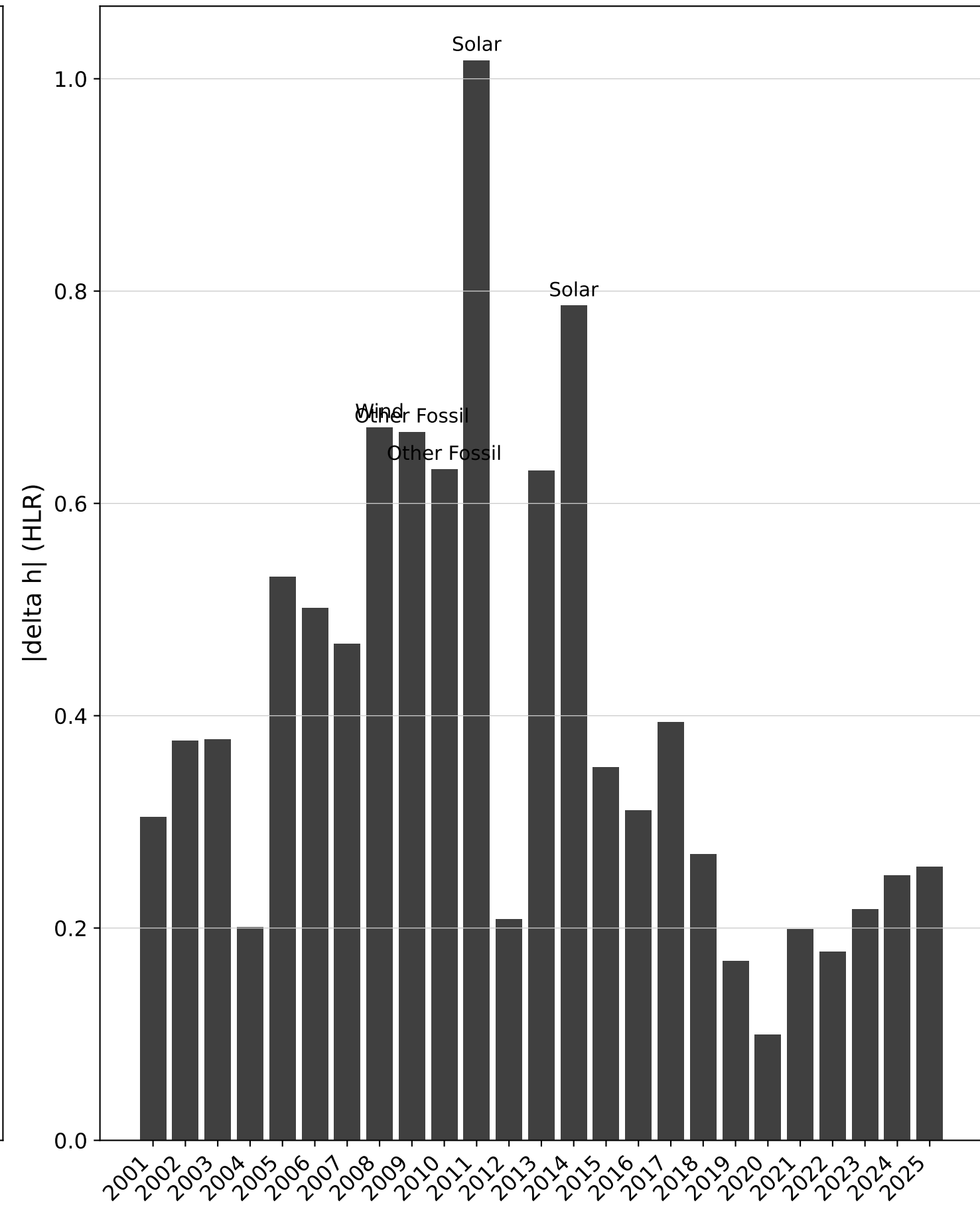


d\_A (HLR)

# HELMSMAN FREQUENCY



# HELMSMAN DISPLACEMENT PER YEAR



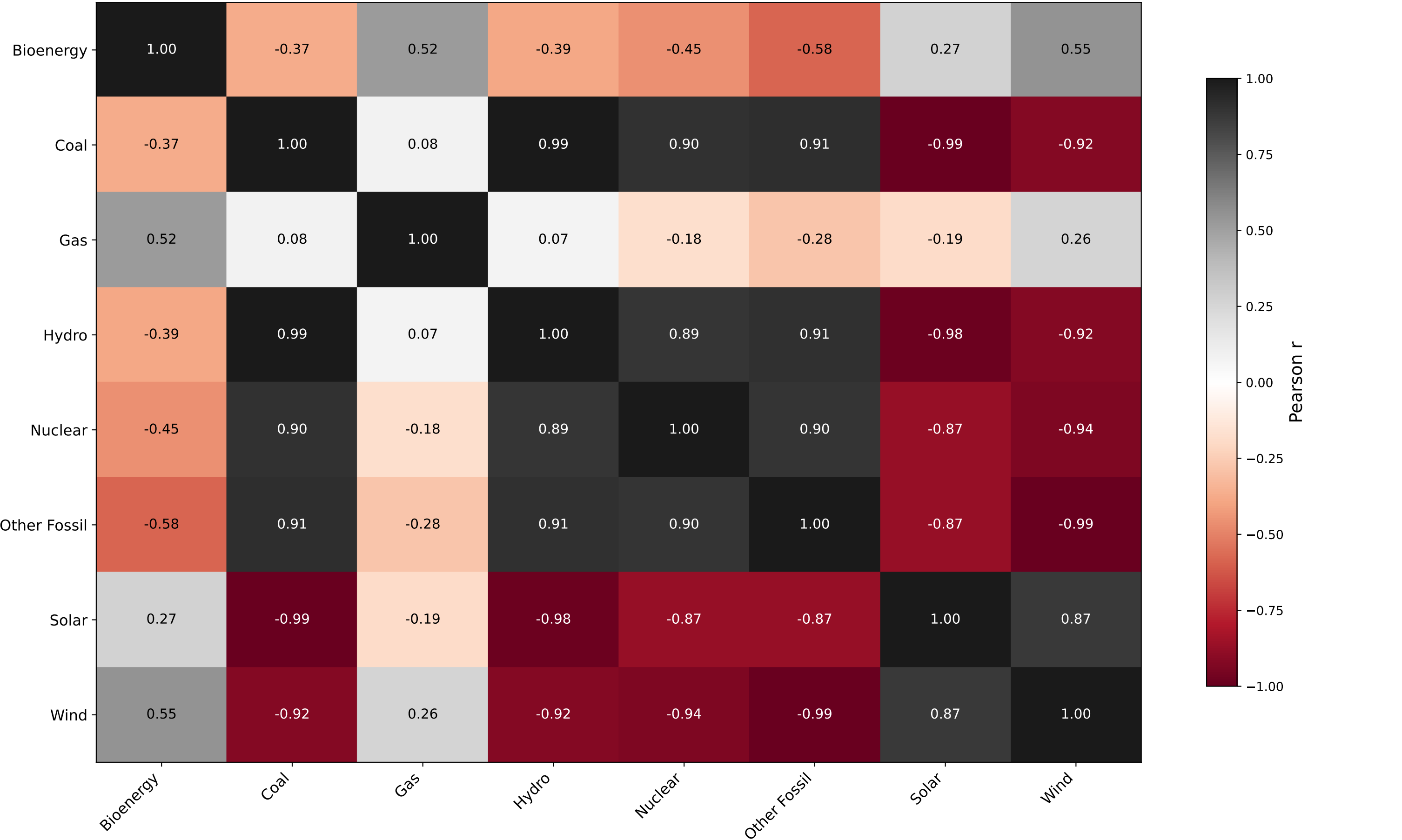
STAGE 2 — Pairwise Divergence Ranking (Top 15)

| Rank | Year A | Year B | d_A (HLR) | Angle (deg) | Dominant Carrier |
|------|--------|--------|-----------|-------------|------------------|
| 1    | 2000   | 2025   | 9.0822    | 79.0        | Solar            |
| 2    | 2001   | 2025   | 8.8101    | 77.4        | Solar            |
| 3    | 2000   | 2024   | 8.8065    | 74.4        | Solar            |
| 4    | 2000   | 2023   | 8.5979    | 70.8        | Solar            |
| 5    | 2001   | 2024   | 8.5376    | 72.8        | Solar            |
| 6    | 2002   | 2025   | 8.5027    | 75.8        | Solar            |
| 7    | 2003   | 2025   | 8.4674    | 75.6        | Solar            |
| 8    | 2004   | 2025   | 8.3657    | 75.2        | Solar            |
| 9    | 2000   | 2022   | 8.3653    | 66.9        | Solar            |
| 10   | 2001   | 2023   | 8.3338    | 69.4        | Solar            |
| 11   | 2002   | 2024   | 8.2356    | 71.4        | Solar            |
| 12   | 2000   | 2021   | 8.2141    | 64.5        | Solar            |
| 13   | 2003   | 2024   | 8.2025    | 71.2        | Solar            |
| 14   | 2005   | 2025   | 8.1607    | 74.9        | Solar            |
| 15   | 2004   | 2024   | 8.1021    | 70.8        | Solar            |

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

| Rank | Year A | Year B | Year C | Area (HLR^2) | Imbalance |
|------|--------|--------|--------|--------------|-----------|
| 1    | 2002   | 2009   | 2025   | 12.2454      | 2.17      |
| 2    | 2003   | 2009   | 2025   | 12.2205      | 2.17      |
| 3    | 2004   | 2009   | 2025   | 12.1074      | 2.18      |
| 4    | 2001   | 2009   | 2025   | 11.8897      | 2.24      |
| 5    | 2000   | 2009   | 2025   | 11.7785      | 2.25      |
| 6    | 2002   | 2009   | 2024   | 11.7128      | 2.10      |
| 7    | 2003   | 2009   | 2024   | 11.6863      | 2.10      |
| 8    | 2004   | 2009   | 2024   | 11.5763      | 2.11      |
| 9    | 2001   | 2009   | 2024   | 11.3801      | 2.17      |
| 10   | 2000   | 2009   | 2024   | 11.2806      | 2.18      |

STAGE 3 — Carrier CLR Trajectory Correlation



# NAVIGATION SUMMARY — All Stage 1 Metrics by Year

| Year | Hs     | Ring | E_metric | omega(deg) | d_A    | Helmsman     | kappa_max |
|------|--------|------|----------|------------|--------|--------------|-----------|
| 2000 | 0.6641 | Hs-4 | 82.7     | 0.00       | 0.0000 | ---          | 67780     |
| 2001 | 0.6543 | Hs-4 | 79.2     | 2.59       | 0.4509 | Solar        | 49360     |
| 2002 | 0.6627 | Hs-4 | 75.1     | 3.35       | 0.5644 | Solar        | 33080     |
| 2003 | 0.6720 | Hs-4 | 74.7     | 3.07       | 0.4638 | Nuclear      | 31843     |
| 2004 | 0.6591 | Hs-4 | 73.2     | 1.90       | 0.2977 | Bioenergy    | 27541     |
| 2005 | 0.6718 | Hs-4 | 69.8     | 5.54       | 0.8427 | Bioenergy    | 31253     |
| 2006 | 0.6832 | Hs-4 | 66.3     | 5.69       | 0.8447 | Other Fossil | 31841     |
| 2007 | 0.6897 | Hs-4 | 64.0     | 3.98       | 0.5782 | Other Fossil | 29832     |
| 2008 | 0.6619 | Hs-4 | 59.6     | 6.82       | 0.9747 | Wind         | 23305     |
| 2009 | 0.6574 | Hs-4 | 52.8     | 7.19       | 1.0426 | Other Fossil | 13267     |
| 2010 | 0.6289 | Hs-4 | 43.0     | 6.06       | 1.0212 | Other Fossil | 6010      |
| 2011 | 0.6306 | Hs-4 | 32.9     | 7.71       | 1.1620 | Solar        | 1806      |
| 2012 | 0.6019 | Hs-4 | 31.6     | 3.85       | 0.3967 | Other Fossil | 1389      |
| 2013 | 0.5936 | Hs-4 | 26.7     | 5.77       | 0.7126 | Solar        | 649       |
| 2014 | 0.5632 | Hs-4 | 21.8     | 8.13       | 0.8546 | Solar        | 246       |
| 2015 | 0.5220 | Hs-4 | 19.3     | 4.64       | 0.4594 | Solar        | 159       |
| 2016 | 0.4878 | Hs-3 | 16.4     | 4.05       | 0.4491 | Solar        | 117       |
| 2017 | 0.4654 | Hs-3 | 14.7     | 6.15       | 0.4775 | Solar        | 118       |
| 2018 | 0.4492 | Hs-3 | 14.1     | 5.00       | 0.3405 | Solar        | 127       |
| 2019 | 0.4262 | Hs-3 | 13.8     | 3.98       | 0.2618 | Other Fossil | 143       |
| 2020 | 0.4068 | Hs-3 | 13.4     | 2.52       | 0.1752 | Bioenergy    | 145       |
| 2021 | 0.3883 | Hs-3 | 13.0     | 4.55       | 0.2936 | Wind         | 150       |
| 2022 | 0.3699 | Hs-3 | 12.1     | 4.21       | 0.2888 | Solar        | 127       |
| 2023 | 0.3596 | Hs-3 | 11.7     | 4.58       | 0.2798 | Solar        | 123       |
| 2024 | 0.3303 | Hs-3 | 11.4     | 4.85       | 0.2907 | Solar        | 113       |
| 2025 | 0.3017 | Hs-3 | 11.7     | 5.00       | 0.2995 | Solar        | 118       |